

Impact of geomagnetic storms on the US electrical grid (2015-2023)

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Abstract:

Both sunspots and solar flares are phenomena caused by changes in the magnetic field of the sun. Observations of these events reveal magnetic activity on the sun, which can in turn impact Earth's magnetic field and interfere with the functioning of radio communication, satellites, and the electrical grid. In light of the economic and practical problems associated with the failure of electrical equipment, it is important to understand the extent to which geomagnetic activity affects the electrical grid. Schriver & Mitchell (2013) found that 4% of disturbances in the US power grid from 1992-2010 are attributable to strong geomagnetic activity. In this research, we test whether there is such a relationship using more recent data on electrical grid disturbances from the Department of Energy from 2015-2023, and measures of geomagnetic activity, namely, the KP Index, Solar Flare Index, and sunspot numbers. Contrary to the findings of Schriver & Mitchell, we find that there is very little correlation between measures of geomagnetic activity and electrical grid disturbances in the US, suggesting that the way disturbances in the electrical grid are recorded have changed, that electrical grid technology has improved to a point that the impact of geomagnetic activity on equipment is minimal, or that geomagnetic activity, in fact, doesn't have a noticeable impact on the functioning of the electrical grid in the first place. Further research on this topic is necessary to discern the cause of this contradiction, and to confirm whether sunspots and solar flares have any correlation with changes to the function of the electrical grid.

Background Information:

- Sunspots are cool, dark spots on the sun caused by changes in its magnetic field, which follows an eleven year cycle correlated with the reversal of the sun's magnetic poles.
- Such changes can also cause solar flares, which are bursts of electromagnetic radiation emitted from the sun (Figure 1).
- Solar magnetic activity, which can be indirectly measured through the occurrences of these two phenomena, influences magnetic activity in the Earth's atmosphere.
- The intensity of solar magnetic activity (solar storms) and its effects on the Earth's atmosphere can be measured in different ways, including the solar flare index, sunspot number, and the KP-index.
- Solar storms can interfere with the transmission of electromagnetic waves on Earth, having the potential to cause costly disturbances in the electrical grid and damages and failures to electrical equipment.
- Oughton et al. (2016) estimated the potential losses associated with catastrophic solar storms to be \$0.5-\$2.7 trillion.

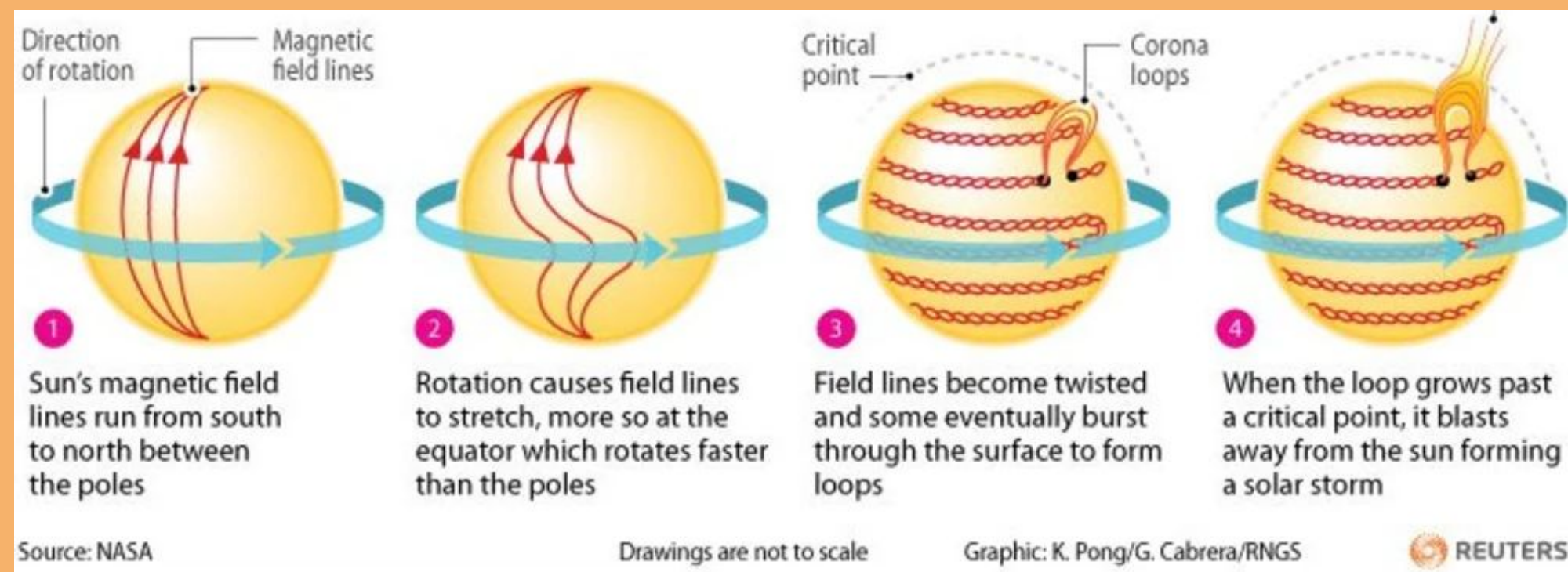


Figure 1: Conceptual figure showing the process of solar flare formation through changes in the sun's magnetic field

Methods and Data:

- National Oceanic and Atmospheric Administration**
 - Daily solar flare index recordings in the Northern Hemisphere (2015-2023).
- US Department of Energy**
 - Data on electrical grid disturbances in the US (2015-2023).
- Helmholtz-Zentrum Potsdam - Deutsches GeoForschungsZentrum GFZ**
 - Three hour measurements of KP-index (2015-2023).
- Solar Influences Data Analysis Center**
 - Daily sunspot number data (2015-2023).
- RStudio & Excel**
 - Packages used: kza, openxlsx, TSA, gsheets, lubridate, and ggplot2
 - Imported data from Excel spreadsheet
 - Created plots and graphs of numerical data
 - Moving average using KZ filter
 - Computed correlation indices
 - Performed chi-square test of independence

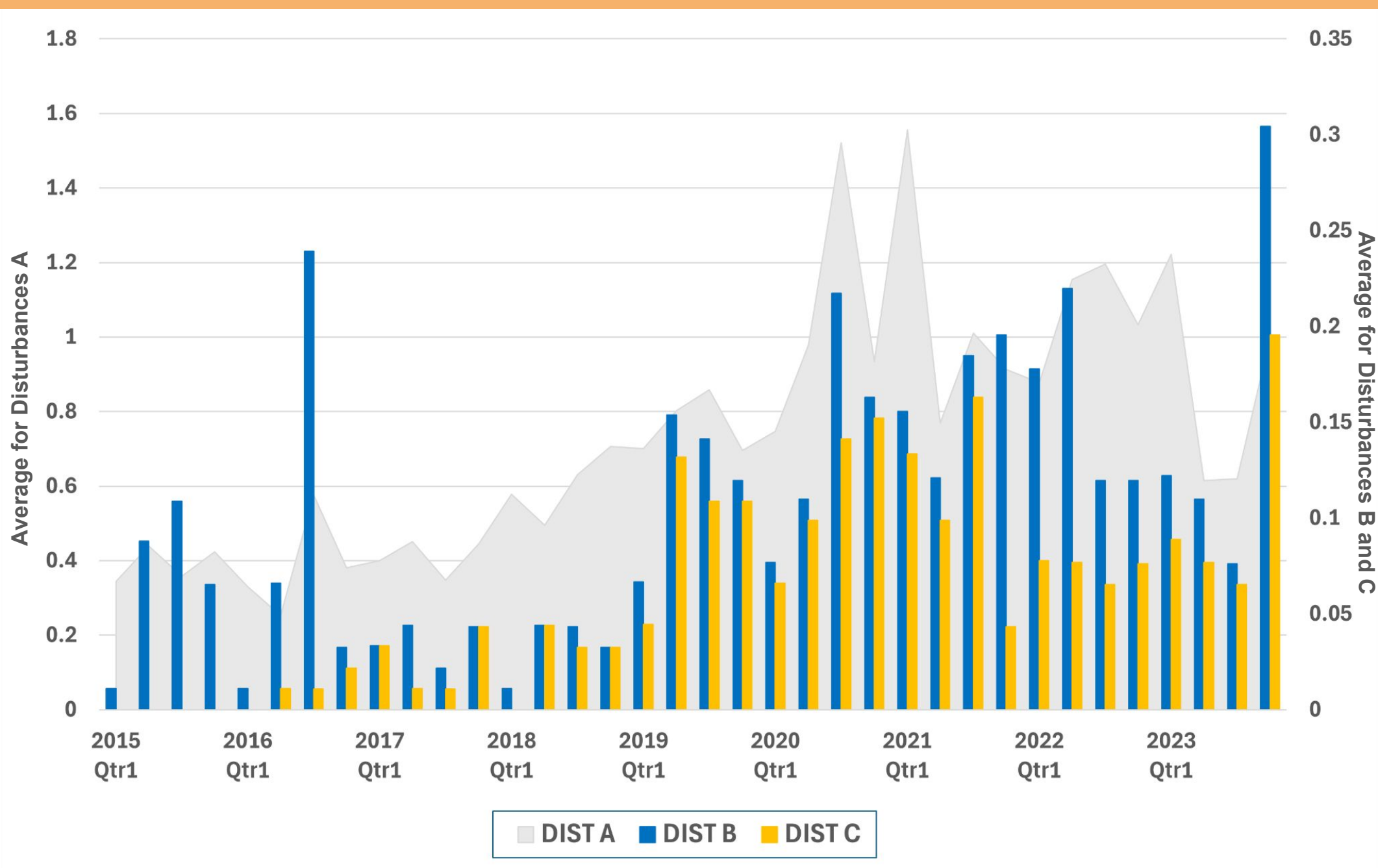


Figure 2: Quarterly averages of electrical grid disturbances, 2015-2023

Source: US Department of Energy

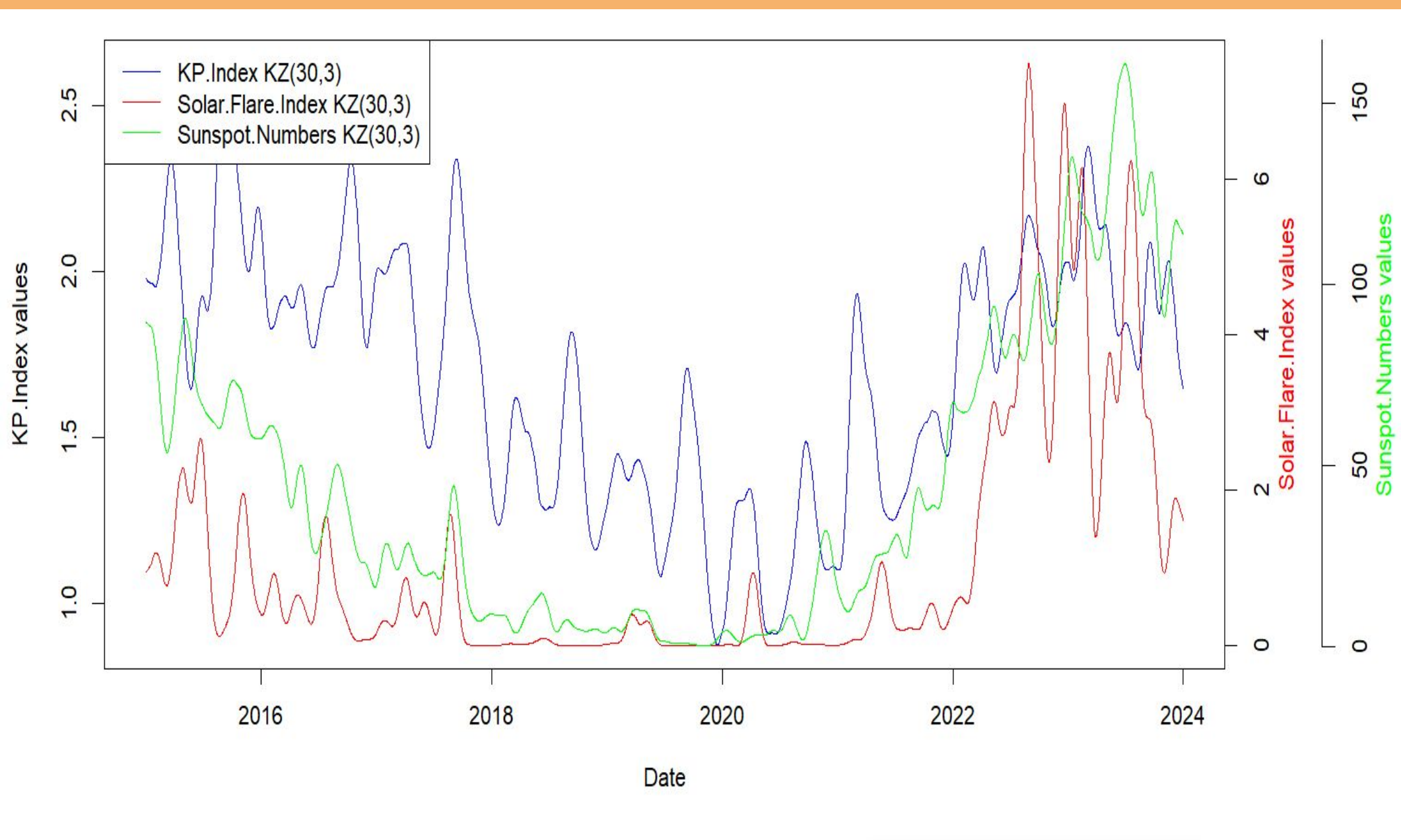


Figure 3: Indices of solar magnetic activity, 2015-2023 (SFI, KP: left axis, inverted; SN: right axis)

Sources: SFI: NOAA, KP: Helmholtz-Zentrum Potsdam, SN: Solar Influences Data Analysis Center

Results:

- Found no correlation indices between alternative daily measures of solar magnetic activity (SN: Sunspot numbers; KP: average KP-index; SFI: Solar flare index) and three different measures of daily counts of electrical grid disturbances:
 - DIST A: all electrical grid disturbances reported to DOE from 2015-2023 with form OE-417.
 - DIST B: disturbances in DIST A plausibly related to geomagnetic activity.
 - DIST C: disturbances in DIST B related to transmission interruptions or failure of electrical equipment.
- The frequency of electric grid disturbances was found to be 1.6% to 5.3% higher on days when $SFI \geq 10$ depending on the definition of disturbances, but these differences are not statistically significant.
- We could not reject the null hypothesis that electric grid disturbances are independent from value of the solar flare index, KP Index, and sunspot numbers.
 - Chi-square test of independence based on the threshold value $SFI=2$
 - Result holds for all measures of disturbances listed above

Table 1 - Correlation Matrix

	KP	SFI	SN	DistA	DistB	DistC
KP	1.000	0.077	0.221	0.003	0.011	-0.007
SFI		1.000	0.393	0.042	0.015	0.024
SN			1.000	0.005	0.047	0.015
DistA				1.000	0.350	0.275
DistB					1.000	0.779
DistC						1.000

Table 2 - Test of statistical independence solar activity and electrical grid disturbance

	Dist A		Dist B		Dist C	
	(a)	(b)	(a)	(b)	(a)	(b)
SFI ≥ 2	184	223	38	369	23	384
SFI<2	1228	1652	275	2605	163	2717
	$\chi^2 = 0.5482$		$\chi^2 = 0.0168$		$\chi^2 = 0.0000$	
SFI ≥ 10	39	42	9	72	7	74
SFI<10	1373	1833	304	2902	179	3027
	$\chi^2 = 0.5209$		$\chi^2 = 0.2201$		$\chi^2 = 1.3062$	

(a): count of days when a disturbance was recorded
(b): count of days when no disturbances were recorded

Source: Authors' elaboration

Discussion/Conclusion:

- Using data from the North American Electric Reliability Corporation (NERC) and the Department of Energy (DOE), Schrijver and Mitchell (2013) found that solar storms caused a statistically significant 4% increase in the daily frequency of electrical grid disturbances between 1992 and 2010.
- Our research sought to validate their results by testing this relationship using more recent DOE data on grid disturbances and various measures of solar magnetic activity for the period 2015-2023.
- We found that grid disturbances were more frequently observed during days of more intense solar activity ($SFI \geq 10$) but could not reject the null hypothesis that daily frequencies of grid disturbances are statistically independent from the intensity of solar flares.
- We found no effect of moderate levels of solar magnetic activity ($SFI \geq 2$) on the frequency of grid disturbances.
- The differences between our results and those in Schrijver and Mitchell (2013) may be attributable to various causes, including (a) changes in the procedures according to which electrical grid disturbances are reported to the DOE and (b) changes in technology, management practices, and the ability to forecast solar storms which have allowed electric utilities to reduce their impacts on the grid.

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Credit Authorship Contribution Statement:

Mazzoleni A.: writing - Abstract, Results, Discussion/Conclusion, References; data analysis - Table 1, Table 2, Figure 2; Raghavendra Rao I.: writing - Background Information, Methods and Data; data analysis - RStudio coding, Table 1, Figure 1, editing; Marsellos A.E.: supervision, guidance, RStudio coding, editing.

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